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Bihar Engineering University, Patna

End Semester Examination - 2023 (Semester IV)

Time: 03 Hours

Code: 104405 ***Subject: Semiconductor Physics and Devices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Write the short answer of the following (Any seven question only): **[2x7=14]**

(a) Define the term drift velocity.

(b) Define Fermi energy level.

(c) Define doping process.

(d) Why is the width of the base region kept very small as compared to other regions in BJT's?

(e) Write the difference between Zener and Avalanche breakdown.

(f) What do you mean by solid state?

(g) Define the effective mass and express it in terms (*E*, K).

(h) Define conductivity and mobility.

(i) Write two differences between BJT and FET.

(j) Define the varactor diode.

Q.2 Solve both questions : **[14]**

(a) Explain (i) Solar Cell (ii) LED (iii) LASER diode.

(b) Discuss the working principle of SCR. How it is different from UJT?

Q.3 Solve both questions : **[14]**

(a) What is MOS Capacitor? Draw its C-V Characteristic.

(b) Explain the operation of enhancement mode MOSFET with neat diagram and compare it with JFET.

Q.4 Solve both questions : **[14]**

(a) Discuss the punch-through breakdown in BJT. Write the expression for the punch though voltage.

(b) Distinguish between the different types of transistor configurations with necessary circuit diagrams.

Q.5 Solve both questions : **[14]**

(a) With a neat diagrams explain the working of a PN junction diode in forward bias and reverse bias and plot the V-I characteristics.

(b) Discuss about the effects of temperature on PN junction diode's Forward and Reverse characteristics.

Q.6 Solve both questions : **[14]**

(a) What is Zener diode. Explain V-I characteristics of Zener diode.

(b) When a transistor is used as a switch, in which region of output characteristics it is operated? Discuss with reason.

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Q.7 Solve this question : **[14]**

What is intrinsic semiconductor and explain the formation extrinsic semiconductors through doping?

Q.8 Solve both questions : **[14]**

(a) Derive the expression for current generated due to drifting of charge carriers in semiconductors in the presence of electric field.

(b) Derive Einstein's relation in semiconductors?

Q.9 Write Short notes ver any two of the following: **[7x2=14]**

(a) charge carriers generation and recombination

(b) UJT relaxation Oscillator

(c) Schottky diode

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